DEDAN KIMATHI UNIVERSITY OF TECHNOLOGY

YEAR III SEMESTER I ORDINARY UNIVERSITY EXAMINATION FOR THE BACHEROR OF SCIENCE IN INFORMATIONB TECHNOLOGY

CIT 3104: COMPUTER GRAPHICS

DATE: 10TH DECEMBER 2019 **TIME: 1400 – 1600 HOURS**

Instructions: Answer Question 1 and Any Other Two.

Question 1: (30 Marks)

a) Explain the three basic phases of the process of creating 3D computer. **(3mks)**.

b) The standard interactive graphics device today is called a raster display.

i) Explain the two types of raster displays. **(4mks)**

ii) Explain the functions of the following components in the operation of raster displays. **(4mks)**
   a) Frame buffer.
   b). Video controller

iii) Draw a well labelled diagram to illustrate how a simple raster graphics system works. **(4mks)**

c) Individual graphic points can be generated by using the primitive type GL_POINTS. By default, a point is drawn as a single pixel which is just barely visible. Which OpenGL function can be used to increase the size of a point? **(1mk)**

d) A direct-view storage tube (DVST) stores the graphic information as a charge distribution just behind the phosphor-coated screen. Outline and explain the purpose of the two electron guns used in a DSTV. **(2mks)**

e) Explain the general syntax for rendering both a solid and a wire sphere cantered at the modelling coordinate's origin of a specified radius and subdivided around the Z axis into slices and along the Z axis into stacks. **(4mks)**

f) Write a program that uses a call back function to render a grey window of size *500 by 500* positioned at the centred of the screen. **(6mks)**

g) Giving an example of each, differentiate between interactive and passive computer Graphics. **(2mks)**

Question 2: (20 marks)

a) Animation adds to graphics the dimension of time which vastly increases the amount of information to be transmitted.

i. Differentiate between computer assisted and computer generated animations. **(4mks)**

ii. In the architecture of raster displays, the pixel data is stored in the frame buffer, which is repeatedly read by the digital-to-analogue converter (DAC) to control the intensity of the electron beam as it sweeps across the screen, one scan-line at a time. Using a well labelled diagram explain why double buffering is required in this type of displays. **(5mks)**

iii. ***glutSwapBuffers()*** function is important in rendering animated graphics in OpenGL. Briefly explain its function. **(3mks)**

b) Explain the functions of the primary and the flood guns in a DVST system. **(2mks)**

c) Explain the purpose of each of the following OpenGl functions. **(6mks)**
i) *glutInit (&argc, argv*);

ii) *glutInitWindowSize();*

iii) *glFlush();*

Question 3: (20 marks)

a) Identify and clearly explain the difference between the three geometric primitives for generating lines in OpenGL. **(6mks)**

b) Indicate the OpenGL type definition for each of the following data types. **(5mks)**

i) 8-bit integer
ii) 32- bit integer
iii) 16-bit unsigned integer
iv) 64-bit floating-point
v) 8-bit unsigned integer

c) Using the glutSwapBuffers() function to ensure smooth animation, write a complete OpenGL program that renders a rotating wire sphere on the screen. (9Mks)

Question 4: (20 marks)

Write a complete OpenGL program that can be used to render the graphic below on the screen. **(20 mks)**

